ModelIt!

Teacher's Guide

Can Nuclear Fusion Power the World?

Modeling Plasma Physics, Energy Balance, and Tokamak Engineering Challenges

Quick Reference

Grade Level	Grade 12L3
NGSS Standard	HS-PS1-8
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS1-8

Develop models to illustrate the changes in the composition of the nucleus of the atom and the energy released during the processes of fission, fusion, and radioactive decay.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a multi-component model to simulate the physics and engineering requirements of nuclear fusion, testing how changes in plasma parameters, magnetic confinement, and materials affect the feasibility of net energy production.
Disciplinary Core Idea	**PS1.C Nuclear Processes** — Students model how the fusion of light nuclei releases energy through mass-energy conversion (E=mc2), analyzing the conditions required to overcome Coulomb repulsion and achieve sustained fusion reactions.
Crosscutting Concept	**Energy and Matter** — Students trace the flow of energy from magnetic and heating input through plasma fusion reactions to neutron kinetic energy, thermal conversion, and electrical output, quantifying the energy balance at each stage.

CAST Testing Alignment

MULTIPLE CHOICE:

A fusion experiment achieves a plasma temperature of 150 million degrees C and adequate density, but confinement time is limited to 5 seconds due to plasma instabilities. Based on the fusion model, why does this result in Q = 0.33 rather than Q > 1, and which parameter improvement would have the greatest impact?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why ITER's achievement of Q = 10 — if successful — would be a historic milestone but still insufficient for commercial fusion power. Reference at least three engineering challenges that must be solved between a Q = 10 demonstration and a power plant generating electricity on the grid.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Nuclear fusion is the process that powers every star in the universe, converting hydrogen into helium and releasing energy according to Einstein's E = mc2. A single gram of fusion fuel (deuterium-tritium) releases as much energy as 8 tons of oil, with no carbon emissions, no long-lived radioactive waste, and fuel abundant enough to power civilization for billions of years. The physics of fusion has been understood since the 1920s. The engineering challenge of replicating stellar conditions on Earth has consumed 70 years and billions of dollars — and may finally be approaching a breakthrough.

The Physics of Fusion

Fusion requires overcoming the Coulomb barrier — the electrostatic repulsion between positively charged nuclei. At room temperature, nuclei repel each other and never get close enough for the strong nuclear force (which operates at femtometer distances) to bind them. At 150 million degrees Celsius, nuclei move fast enough that a small fraction overcome the Coulomb barrier through quantum tunneling and fuse. The deuterium-tritium (D-T) reaction is the easiest to achieve because it has the lowest temperature requirement and highest cross-section: D + T -> He-4 (3.5 MeV) + neutron (14.1 MeV). The helium nucleus (alpha particle) stays in the plasma and helps maintain temperature, while the neutron escapes and carries 80% of the energy to the reactor walls, where it must be captured and converted to heat.

The Lawson Criterion and Triple Product

In 1955, John Lawson calculated the minimum conditions for a fusion reactor to produce more energy than it consumes. The 'triple product' — plasma temperature x density x confinement time — must exceed approximately 3 x 10^21 keV seconds per cubic meter for D-T fusion. This criterion explains why fusion is so hard: increasing temperature makes confinement more difficult (hotter plasma is harder to contain), increasing density creates instabilities, and increasing confinement time requires suppressing plasma turbulence that develops within milliseconds. Current experiments have achieved each parameter individually but struggle to maintain all three simultaneously for practical durations.

Magnetic Confinement and Tokamaks

The tokamak, invented in the Soviet Union in the 1950s, confines plasma using a combination of toroidal (doughnut-shaped) and poloidal (circular cross-section) magnetic fields that create a helical field structure trapping charged particles on closed magnetic surfaces. The plasma floats in a magnetic bottle, never touching any physical wall. ITER, under construction in southern France, is a tokamak designed to produce 500 MW of fusion power from 50 MW of heating input (Q = 10) for pulses of 400-600 seconds. Its superconducting magnets generate fields of 11.8 Tesla using niobium-tin superconductor cooled to 4.2 Kelvin. Next-generation designs from Commonwealth Fusion Systems use high-temperature superconductors (REBCO tape) that can achieve 20+ Tesla in smaller, cheaper devices.

The Path to Commercial Fusion

Achieving Q > 1 is a physics milestone; building a power plant is an engineering marathon. Commercial fusion requires: sustained Q > 30 (to cover all plant systems, not just plasma heating), continuous operation (ITER runs in pulses), tritium breeding ratio > 1.0 (the reactor must produce more tritium than it consumes), wall materials that withstand years of neutron bombardment, remote maintenance robotics (the reactor interior becomes radioactive), and competitive cost of electricity (currently estimated at $50-100/MWh for first-generation plants). Private companies like Commonwealth Fusion Systems, TAE Technologies, and Helion Energy are pursuing faster timelines than the ITER pathway, aiming for demonstration plants by the early 2030s.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components of the fusion energy system: fuel supply, magnetic field strength, energy input, plasma temperature, plasma density, confinement time, fusion reaction rate, neutron flux, wall material degradation, and net energy output.

E **Establish Relationships**

Students map relationships showing how the triple product determines fusion rate, how neutron flux creates materials challenges independent of plasma physics, and how tritium breeding creates a fuel cycle constraint.

V **Visualize & Model**

Students build a ten-component computational model showing the energy balance of a fusion reactor with competing requirements for plasma performance, materials endurance, and fuel self-sufficiency.

E **Evaluate Outcomes**

Students run current best performance, ITER target, and commercial plant scenarios to quantify the gap between demonstrated fusion performance and the requirements for grid-scale electricity generation.

R **Revise & Extend**

Students add plasma instability modes, superconducting magnet cooling, or blanket module engineering to model the detailed engineering challenges within each major system.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"The Sun fuses 620 million tons of hydrogen every second and will continue for another 5 billion years. If we could replicate even a tiny fraction of that process on Earth, we would have unlimited clean energy forever. Scientists have been trying for 70 years. Today you will understand exactly why it is so hard — and why we might finally be close."*

Watch For

Show cover image. Ask: what powers the Sun? How much energy does fusion release compared to burning coal? (Answer: about 10 million times more per unit mass.) Let the scale of the energy difference sink in.

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling the most ambitious energy technology ever attempted. You will discover why fusion requires temperatures ten times hotter than the Sun's core, why containing plasma is like holding lightning in a bottle, and why '30 years away' might finally be shrinking."*

Watch For

Have students read objectives. Pre-teach the Coulomb barrier: positive nuclei repel each other like same-pole magnets. Only extreme speed (temperature) allows them to overcome this repulsion and fuse. Demonstrate with magnets.

Slide 3: Big Question

3 min

Say: *"One gram of fusion fuel releases as much energy as 8 tons of oil. The fuel is in seawater — essentially unlimited. No carbon emissions. No long-lived waste. So why don't we have fusion power plants yet? The answer is the Lawson criterion — the hardest physics target in energy engineering."*

Watch For

Quick-write: Students write what they think is the hardest part of building a fusion reactor. Share out and catalog answers. Reveal that the answer is 'all of the above, simultaneously.'

Slide 4: LEVER Framework

2 min

Say: *"We will Label the ten components of a fusion energy system, Establish the physics relationships that determine whether fusion produces or consumes energy, Visualize the model, Evaluate three levels of fusion performance, and*

Refine by adding plasma instabilities and engineering subsystems."

Watch For

Preview each LEVER step. Emphasize that fusion physics works — we know how to make nuclei fuse. The challenge is engineering a system that does it efficiently, sustainably, and affordably.

Slide 5: Activity 1: Components

8 min

Say: *"Three inputs define what you have to work with: fuel, magnetic field, and energy for heating. Seven internal variables determine whether the reactor produces or consumes energy. Sort them and identify which components represent the triple product of the Lawson criterion."*

Watch For

Students sort external versus internal components. Identify the Lawson triple product: Plasma Temperature x Plasma Density x Confinement Time = Fusion Reaction Rate. Discuss: why must all three be achieved simultaneously?

Slide 6: Activity 2: Connections

8 min

Say: *"Magnetic field strength improves confinement. Higher temperature increases fusion rate but makes confinement harder. Neutron flux damages walls but carries the energy we want to capture. These tensions between competing requirements are why fusion engineering is so challenging."*

Watch For

Students draw arrows showing relationships. Identify the key tension: temperature must be maximized for fusion but makes confinement harder. Discuss how magnetic field strength mediates this tension.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run three scenarios: current best performance (Q = 0.33), ITER's target (Q = 10), and a commercial power plant (Q > 30). Watch the energy balance and identify where each scenario falls relative to the Lawson criterion."*

Watch For

Students run all three scenarios and compare Q factors. Key discussion: why is Q = 10 a physics triumph but an engineering starting point? What additional factors beyond Q must a commercial plant address?

Slide 8: Discoveries

5 min

Say: *"Your model reveals four critical truths: the Lawson criterion demands simultaneous extremes, confinement time is the current bottleneck, neutron damage is the hidden engineering challenge, and tritium self-sufficiency is a non-negotiable constraint. Fusion is an engineering problem, not a physics problem — and that is actually good news."*

Watch For

Class discussion of discoveries. Ask: why is it good news that fusion is an engineering problem rather than a physics problem? (Because engineering problems have solutions — they require money, time, and ingenuity, not new physics.)

Slide 9: STEM Challenge

12 min

Say: *"Design the specifications for a 1 GW fusion power plant that could come online by 2045. Your model data defines the physics requirements. Your job is to solve the engineering: magnets, materials, tritium, and cost. Every specification must be justified by model evidence."*

Watch For

Groups design power plant specifications covering plasma parameters, magnet technology, wall materials, tritium breeding, and cost estimates. Present and evaluate: which design is most feasible? Which is most innovative? What is the biggest unsolved challenge?

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Deuterium-Tritium Fuel Supply, Magnetic Field Strength, Energy Input (Heating)
- **INTERNAL:** Plasma Temperature, Plasma Density, Confinement Time, Fusion Reaction Rate, Neutron Flux on Walls, Wall Material Degradation, Net Energy Output (Q Factor)

Relationships

- **Magnetic Field Strength to Confinement Time:** POSITIVE (+) — Stronger magnetic fields compress plasma more tightly and resist plasma instabilities more effectively, increasing the duration for which the plasma remains confined and hot enough for fusion.
- **Plasma Temperature to Fusion Reaction Rate:** POSITIVE (+) — Higher plasma temperature means nuclei move faster and more frequently overcome the Coulomb barrier, exponentially increasing the fusion reaction rate up to an optimal temperature of approximately 15 keV for D-T reactions.
- **Plasma Density to Fusion Reaction Rate:** POSITIVE (+) — Higher plasma density means more fuel nuclei per unit volume, increasing the probability of collisions and fusion events. The fusion reaction rate scales with the square of density.
- **Confinement Time to Net Energy Output:** POSITIVE (+) — Longer confinement time allows more fusion reactions to occur before the plasma loses its energy, directly increasing the total fusion energy produced per plasma pulse and improving the Q factor.
- **Fusion Reaction Rate to Neutron Flux on Walls:** POSITIVE (+) — Higher fusion reaction rates produce more 14.1 MeV neutrons per second. Since neutrons are electrically neutral and unaffected by magnetic fields, they stream directly to the reactor walls, increasing the bombardment intensity.
- **Neutron Flux on Walls to Wall Material Degradation:** POSITIVE (+) — Higher neutron flux causes more atomic displacements, helium bubble formation, and nuclear transmutation in wall materials per unit time, accelerating structural degradation and reducing the operational lifetime of inner reactor components.
- **Energy Input to Plasma Temperature:** POSITIVE (+) — More heating power from neutral beams, radio-frequency waves, and ohmic current raises the plasma temperature. Achieving fusion-relevant temperatures of 150 million degrees requires tens of megawatts of sustained heating.
- **Wall Material Degradation to Net Energy Output:** NEGATIVE (-) — As wall materials degrade, they release impurities into the plasma that radiate energy and cool it, reducing fusion performance. Degraded walls also reduce tritium breeding efficiency and eventually require reactor shutdown for replacement, reducing overall energy output.

Key Discoveries

- The Lawson criterion reveals why fusion is so difficult — achieving the necessary triple product of temperature, density, and confinement time simultaneously requires solving multiple physics and engineering challenges that work against each other: higher temperature makes confinement harder, higher density creates instabilities, and longer confinement requires more stable plasma
- Confinement time is the current bottleneck — plasma temperature and density are achievable with current technology, but maintaining confinement for hundreds of seconds against plasma instabilities that develop within milliseconds requires extraordinary magnetic field precision and real-time feedback control
- Neutron damage to reactor walls is the hidden engineering challenge — even after achieving net energy gain, the 14.1 MeV neutrons from D-T fusion will degrade structural materials so rapidly that inner wall components must be replaced every few years, creating a materials science challenge that has no solution yet

- Tritium self-sufficiency is a critical constraint — a commercial fusion plant would consume approximately 56 kg of tritium per year, but the world's entire tritium supply is about 25 kg, meaning the reactor must breed its own tritium from lithium blankets with greater than 100% efficiency (tritium breeding ratio > 1.0), which has never been demonstrated

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Fusion is dangerous like nuclear fission — it could melt down"	A fusion reactor cannot melt down because the fusion reaction requires precise conditions that are inherently difficult to maintain. If containment is lost, the plasma instantly cools to below fusion temperature and the reaction stops within milliseconds. There is no chain reaction to control — maintaining fusion IS the challenge, not preventing runaway. The total fuel in a fusion reactor at any moment is less than 1 gram, compared to tens of tons of fuel in a fission reactor. Fusion produces helium (non-radioactive) as its primary waste, not the long-lived radioactive isotopes produced by fission.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We already have unlimited energy from solar and wind, so we don't need fusion"	Solar and wind are excellent and rapidly growing but have fundamental limitations: intermittency (the sun does not always shine, the wind does not always blow), energy density (solar farms require enormous land area), storage (batteries remain expensive for grid-scale seasonal storage), and geographic constraints. Fusion would provide baseload power — reliable, 24/7, weather-independent, high-energy-density electricity — that complements intermittent renewables. A complete clean energy system likely needs both fusion for baseload and renewables for distributed generation.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Fusion is basically solved since NIF achieved ignition in 2022"	NIF's December 2022 result was a historic milestone — the first time a fusion reaction produced more energy than the laser energy delivered to the fuel capsule. However, the lasers themselves required approximately 300 MJ of electrical energy to produce the 2.05 MJ of laser light that triggered the fusion reaction producing 3.15 MJ. The overall system consumed 100x more energy than fusion produced. NIF uses inertial confinement (lasers) rather than magnetic confinement (tokamaks), and was designed for weapons research, not power generation. The result proves the physics works but highlights the enormous engineering gap to commercial power.